UNCONDITIONAL GLOBAL SQUARECLASSES, GENUS-161 RATIONAL-POINT CLASSIFICATION, AND MAP RIGIDITY ON BREMNER'S OCTIC K3 SURFACE

ARGUS OPEN RESEARCH PROJECT

Abstract. We determine the exact set of rational points on the genus-13 curve $H\colon s^2 = G_1(x)$ and on the genus-161 three-square cover $X\colon y_1^2 = G_1(x),\ y_2^2 = G_2(x),\ y_3^2 = G_3(x)$ arising from the second-iterate Category 3 parameterization on Bremner's octic K3 surface for 3×3 magic squares of squares. Without assuming GRH or finiteness of Ш, we certify odd class number $h(K)$ for the degree-18 two-division field $K/\mathbb{Q}$, prove that the twelve archived Simon generators span the complete kernel $\ker(N_{K/k}\colon K(S_K, 2) \to k(S_k, 2))$, and combine this with the Cassels–Tate pairing to establish $\dim_{\mathbb{F}_2} \mathrm{Sel}_2(E/k) = 3$, $\mathrm{rank}\, E(k) = 1$, and $Ш(E/k)[2^\infty] \cong (\mathbb{Z}/2\mathbb{Z})^2$ over the sextic base field k. Together with $p = 23$ Chabauty–Coleman closure across all eight quadratic twists of $C_1\colon y^2 = f(x)$, this proves that $H(\mathbb{Q}) = \{(-1, \pm 896)\}$ and $|X(\mathbb{Q})| = 8$, all lying above the collision locus $x = -1$. Finally, we prove that X has geometric gonality 8 with a unique degree-8 pencil, $\mathrm{Aut}_{\mathbb{Q}}(X) \cong (\mathbb{Z}/2\mathbb{Z})^3$, and admits no nonconstant rational morphism to Bremner's genus-57 cover.

1. Main Mathematical Results

Theorem 1.1 (Unconditional arithmetic classification and rigidity of the Branch-483 cover)**.** *For the sextic field $k = \mathbb{Q}(z)$ ($z^6 - 2z^5 + 3z^4 - 36z^3 + 60z^2 + 32z - 56 = 0$), elliptic curve E/k, genus-13 curve H, and genus-161 cover X:*

(1) *Unconditionally,* $\dim_{\mathbb{F}_2} \mathrm{Sel}_2(E/k) = 3$, $\mathrm{rank}\, E(k) = 1$, *and* $Ш(E/k)[2^\infty] \cong (\mathbb{Z}/2\mathbb{Z})^2$.

(2) *The smooth projective curve $C_1\colon y^2 = f(x)$ satisfies $C_1(\mathbb{Q}) = \{\infty_\pm\}$, $H(\mathbb{Q}) = \{(-1, \pm 896)\}$, and $X(\mathbb{Q})$ consists of the eight affine points above $x = -1$, all mapping to the magic-square collision divisor.*

(3) *The curve X has geometric gonality* 8, $\mathrm{Aut}_{\mathbb{Q}}(X) \cong (\mathbb{Z}/2\mathbb{Z})^3$, $\mathrm{End}_{\overline{\mathbb{Q}}}(J(C_1)) \cong \mathbb{Z}$, *and admits no nonconstant rational map to any curve of genus* 57.

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1243 in the domain of arithmetic geometry and Diophantine equations adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Unconditional 12-dimensional norm-square kernel over the degree-18 two-division field, rank E(k)=1, Sha(E/k)[2^infinity]=(Z/2Z)^2, complete rational points on the genus-13 and genus-161 residual curves, and geometric gonality/automorphism rigidity.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Unrestricted nonexistence of 3x3 magic squares of nine distinct squares

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References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 1243 (arithmetic geometry and Diophantine equations), archived in the accompanying repository package `results/1243/`.